SANDOZ

TFM PROPOSAL

AutoComments

Problem Statement

Management reporting within large organizations relies heavily on written comments that explain the drivers behind financial variances versus prior year, budget, or forecast. These comments are typically produced manually by finance teams based on accounting database line items (e.g., Sales, Core EBIT, TFC) and business dimensions such as geography, product, or market cluster.

This manual commentary process presents several limitations:

- It is time-consuming and requires significant senior expertise to ensure accuracy, consistency, and appropriate business tone.
- The quality and depth of explanations may vary depending on the author, leading to inconsistencies across regions and reporting cycles.
- Analysts often focus on producing the narrative rather than on deeper value-adding analysis or decision support.

The *AutoComments* Master Thesis aims to read financial line items (directly from finance databases) and contextual information, identify the main variance drivers, and generate natural-language comments similar in style and structure to those currently produced by finance professionals. There are similar initiatives the most recent ones from Amazon Insights Agent or DataSage from ByteDance

Proposed Approach

The proposed approach consists of designing and implementing an AI-based

system able to generate explanatory comments for financial variances by analyzing structured financial data. The focus will be on reproducing the logic, structure, and business language typically used in management comments, such as identifying negative or positive drivers, highlighting key products, markets, or regions, and linking operational events to financial outcomes.

The work will follow these main steps:

1. **Data Understanding and Preparation**

Accounting data will be analyzed to understand available dimensions (e.g., geography, product, customer, period) and key financial metrics. Historical datasets containing both numerical data and existing written comments will be used, where available, to establish reference patterns.

2. **Driver Identification Logic**

Analytical rules and/or machine-learning techniques will be developed to detect the main contributors to variances (e.g., largest negative deltas by product or region, structural effects such as phasing or price pressure). This step ensures the generated comments are grounded in explainable financial logic.

3. **Natural Language Generation**

Based on the identified drivers, an AI model will generate structured text comments that mirror real business examples.

4. **Validation and Evaluation**

The output comments will be evaluated against existing manual comments and reviewed with finance stakeholders to assess accuracy, relevance, readability, and consistency.

Technical tools

The project will utilize LLMs in combination with Python and relevant libraries to implement and deploy the models effectively. Optionally we could add a database for adding text-to-SQL capabilities.